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Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment- 7

Lab Title: AI-ML Programs

Naive Bayes — Scenario-Based Coding Questions

This document contains four detailed, scenario-based coding questions designed exclusively around Naive Bayes classifiers. Each question focuses on applying different variants of Naive Bayes (Gaussian, Multinomial, or Bernoulli) to real-world problems. Datasets for each problem are provided, along with step-by-step tasks to be implemented.

Question 1: Spam Email Classification

Scenario:
You are given a dataset of emails labeled as 'spam' or 'not spam'. The features represent word counts or term frequencies (bag-of-words) for a vocabulary of around 1000 words. The goal is to automatically detect spam messages using Naive Bayes.

Dataset link: https://archive.ics.uci.edu/ml/datasets/SMS+Spam+Collection

1. Tasks:
2. 1. Choose the appropriate Naive Bayes variant (e.g., MultinomialNB) and justify your choice.
3. 2. Preprocess the text data by tokenizing, cleaning, and vectorizing it using CountVectorizer or TF-IDF.
4. 3. Train the Naive Bayes model, make predictions on a test set, and compute evaluation metrics such as accuracy, precision, recall, and F1-score.
5. 4. Discuss how your model would handle words that appear in the test set but not in training, and propose smoothing techniques (e.g., Laplace smoothing).
6. 5. (Bonus) Improve model performance using stopword removal, feature selection, or TF-IDF weighting.

Question 2: Predicting Customer Churn (Continuous + Categorical Features)

Scenario:
A telecom company provides you a dataset where each record corresponds to a customer. Features include monthly usage (continuous), number of calls (continuous), contract type (categorical: 'month-to-month', 'one-year', 'two-year'), and whether they have an international plan (binary). The target variable is churn = yes/no. Your task is to predict customer churn using Naive Bayes.

Dataset link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

7. Tasks:

8. 1. Decide how to handle mixed continuous and categorical features under Naive Bayes (e.g., use GaussianNB for continuous, CategoricalNB for discrete).
9. 2. Implement preprocessing — discretize continuous variables or assume Gaussian distribution as needed.
10. 3. Train and evaluate your Naive Bayes model using confusion matrix and AUC score.
11. 4. Compare the Naive Bayes model's performance with a logistic regression baseline model.
12. 5. Discuss which features might violate the independence assumption and how that could affect the model's results.

Question 3: Sentiment Classification of Movie Reviews

Scenario:
You are given a dataset of movie reviews labeled as positive or negative. Your goal is to build a Naive Bayes classifier that predicts the sentiment of a given movie review.

Dataset link: https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews

13. Tasks:
14. 1. Choose a Naive Bayes variant (MultinomialNB or BernoulliNB) and justify your selection.
15. 2. Preprocess text (tokenize, lowercase, remove punctuation, and optionally apply stemming or lemmatization).
16. 3. Convert text into numerical features using CountVectorizer or TF-IDF representation.
17. 4. Train, test, and evaluate the model using metrics such as accuracy, precision, recall, and F1-score.
18. 5. Handle the inclusion of n-grams (e.g., bigrams) as features to improve model context understanding.
19. 6. (Bonus) Compare performance using unigrams vs. bigrams and with/without smoothing techniques.

Question 4: Disease Prediction from Medical Records

Scenario:
You are provided with a medical dataset containing patient records. Features include age (continuous), blood pressure (continuous), cholesterol level (continuous), smoking status (binary), family history (binary), and BMI category (categorical: underweight/normal/overweight). Your goal is to predict whether a patient has a particular disease using Naive Bayes.

Dataset link: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

20. Tasks:
21. 1. Determine how to handle continuous features (GaussianNB) and categorical features (CategoricalNB).
22. 2. Preprocess the data (scaling, encoding, or discretization as required).
23. 3. Train the Naive Bayes model and evaluate its performance using metrics such as accuracy, precision, recall, and F1-score.
24. 4. If some feature distributions are highly skewed (e.g., cholesterol), apply transformations or use non-parametric methods.
25. 5. Discuss how correlated features (e.g., age and BMI) may violate the independence assumption, and suggest mitigation approaches (e.g., feature selection).